SPRING BOOT · COURSE PROJECT

Online Store Backend

Build the backend service for an online shopping platform. Design the data, the API, and the security yourself — the choices are part of the assignment.

> You are building the server side of a small online store. Customers browse products, add them to a cart, and place orders. Administrators manage the catalog. Everything is exposed as a web API — there is no user interface to build. How you model the data and structure your code is up to you.

1 The Scenario

A small business wants a backend for its online shop. It should let people create an account, browse and search products organized into categories, keep a shopping cart, place orders, and leave reviews on products they can see. Staff members need a way to manage the products and categories and to keep an eye on incoming orders.

Your job is to design and build that backend. Decide what information the system needs to store, how the pieces relate, and how the API should be organized so it is clear and easy to use.

2 Who Uses the System

CUSTOMER

- Create an account and log in.
- Browse and search the product catalog and view product details.
- Add products to a personal shopping cart and adjust it.
- Place an order from the cart and view their own order history.
- Leave a review on a product.

ADMINISTRATOR

- Add, update, and remove products and categories.
- Manage product stock and pricing.
- View orders placed in the system.

3 What the System Must Do

Accounts & Access

- People can register and log in.
- Actions are available only to the right kind of user — some things any logged-in customer can do, others only an administrator can do.
- A user should never be able to reach another user's private information.

Products & Categories

- Products belong to categories and can be browsed by category.
- Customers can search and filter the catalog.
- The catalog can be large, so browsing it should return results in a sensible, manageable way rather than everything at once.

Shopping Cart

- Each customer has their own cart.
- Customers can add items, change quantities, and remove items.

Orders

- A customer can turn their cart into an order.
- The order records what was bought, in what quantity, and at what price.
- Customers can see their own past orders.

Reviews

- Customers can leave a review on a product, and reviews are tied to the person who wrote them.

4 Business Rules to Respect

- **Privacy.** A customer can only see and change their own cart and their own orders — never anyone else's, even if they try.
- **Stock.** An order cannot be placed for more of a product than is actually available. If it can't be fulfilled, nothing about it should be half-saved.
- **Permissions.** Only administrators can change the catalog. A regular customer must not be able to add, edit, or delete products or categories.
- **Good behavior.** The API should respond sensibly when something goes wrong — bad input, a missing item, or an action the user isn't allowed to take — with a clear, appropriate response rather than a crash.

5 What to Hand In

Deliverables

- The complete, working project source code.